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Dissecting barrier

Abstract

The present subject matter provides a dissecting element that comprises a flat body having an upper side and a bottom side and opposite edges; a hinge on the bottom side wherein the hinge is connected to a surface, wherein the dissecting element is configured to move between an erected state in which the flat body is in an upright position and a horizontal state in which the flat body is adjacently lying on the surface. A plurality of dissecting elements can be provided on a portable ramp and several dissecting elements or ramps establish a barrier system. A method is also provided to prevent passage of vehicles using the dissecting elements.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This patent application is a U.S. National Phase filing under 35 U.S.C. § 371 of co-pending PCT Patent Application No. PCT/IB2020/051701, filed Feb. 27, 2020, which is based upon and claims the priority of U.S. Provisional Patent Application Ser. No. 62/811,576, filed Feb. 28, 2019, each of is incorporated herein by reference in its entirety.

FIELD

(1) The present subject matter relates to barriers. More particularly, the present subject matter relates to barriers configured to dissect vehicles.

BACKGROUND

(2) There are occasions when there is a need to block passage of vehicles with a barrier, by any means, including dissecting the vehicle when attempting to pass across the barrier, in order to absolutely prevent passage of the barrier by the vehicle.

SUMMARY

(3) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this subject matter belongs. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present subject matter, suitable methods and materials are described below. In case of conflict, the patent specification, including definitions, will control. In addition, the materials, methods, and examples are illustrative only and not intended to be limiting.

(4) According to one aspect of the present subject matter, there is provided a dissecting element comprising: a flat body having an upper side and a bottom side and opposite edges; a hinge on the bottom side wherein the hinge is connected to a surface, wherein the dissecting element is configured to move between an erected state in which the flat body is in an upright position and a horizontal state in which the flat body is adjacently lying on the surface.

(5) According to another embodiment, at least one of the opposite edges is sharp.

(6) According to another embodiment, the surface is a portable ramp.

(7) According to another embodiment, the surface is a ground onto which vehicles pass.

(8) According to another aspect of the present subject matter, there is provided a dissecting barrier unit comprising: at least one dissecting element having a flat body with opposite edges; a ramp onto which said at least one dissecting unit is pivotally connected; wherein the at least one dissecting element is configured to move between an erected state in which the flat body is in an upright position and a horizontal state in which the flat body is adjacently lying on the ramp.

(9) According to another embodiment, the at least one dissecting element is pivotally connected to the ramp by a hinge.

(10) According to another embodiment, the ramp is configured to be anchored to a surface.

(11) According to another embodiment, the ramp comprises at least one anchoring element configured to facilitate anchoring the ramp to the surface.

(12) According to another embodiment, the anchoring element is in a form of a hole, configured to accommodate a sticking element, that is configured to be stuck in the surface.

(13) According to another embodiment, the ramp is permanently anchored to the surface.

(14) According to another embodiment, the ramp is reversibly anchored to the surface.

(15) According to another embodiment, the ramp is configured to be connected to at least one other ramp.

(16) According to another embodiment, the ramp is configured to permanently be connected to the at least one other ramp.

(17) According to another embodiment, the ramp is configured to reversibly be connected to the at

least one other ramp.
(18) According to another embodiment, the ramp further comprising at least one connecting element for connecting the ramp to the at least one other ramp.
(19) The dissecting barrier unit of claim **15**, wherein the connecting element is a tube-like element attached beneath the ramp and configured to accommodate a shaft-like element configured to be inserted into the tube-like element of one ramp and a tube-like element of an adjacent ramp, thus connecting the ramps and holding them together.
(20) According to another embodiment, the ramp is configured to allow passage of a vehicle over the ramp when the dissecting elements are in a horizontal state.
(21) According to another embodiment, the ramp is substantially flat.
(22) According to another embodiment, the ramp is substantially curved.
(23) According to another embodiment, a degree of curvature of the ramp is such that it allows free passage of a vehicle over the ramp when the dissecting elements are in a horizontal state.
(24) According to another embodiment, the flat body has an upper side that is wide and lower side that is relatively narrow.
(25) According to another embodiment, the dissecting element is configured to partially dissect an approaching vehicle through impact with the edges of the flat body.
(26) According to another embodiment, at least part of the edges of the dissecting element is sharp, having a blade-like shape.
(27) According to another embodiment, the dissecting element further comprises a stopper configured to stop movement of the dissecting element and firmly hold the dissecting element in the erected state.
(28) A dissecting barrier system is also provided that comprises a plurality of dissecting elements in accordance with embodiments of the subject matter.
(29) A dissecting barrier system is also provided that comprises a plurality of dissecting barrier units with embodiments of the subject matter.
(30) According to another embodiment, at least two of the dissecting barrier units are connected one to the other.
(31) According to another embodiment, each one of the dissecting elements is independently controlled.
(32) According to another embodiment, each one of the dissecting barrier units is adapted to be in an erected state or in horizontal state.
(33) According to another embodiment, each one of the dissecting barrier units is independently controlled.
(34) According to another embodiment, the dissecting barrier units are remotely controlled.
(35) According to another embodiment, the dissecting elements are remotely controlled.
(36) A method of blocking vehicles is also provided in accordance with another embodiment comprises providing a dissecting element, and positioning the dissecting element with the opposite edges directed towards a direction from which the vehicles are approaching.
(37) Another method of blocking vehicles is also provided in accordance with another embodiment comprises providing a dissecting barrier unit and positioning the at least one dissecting element with the opposite edges directed towards a direction from which the vehicles are approaching.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Embodiments are herein described, by way of example only, with reference to the accompanying drawings. With specific reference now to the drawings in detail, it is stressed that the particulars shown are by way of example and for purposes of illustrative discussion of the

preferred embodiments, and are presented in the cause of providing what is believed to be the most useful and readily understood description of the principles and conceptual aspects of the embodiments. In this regard, no attempt is made to show structural details in more detail than is necessary for a fundamental understanding, the description taken with the drawings making apparent to those skilled in the art how several forms may be embodied in practice.
(2) In the drawings:
(3) FIG. **1** schematically illustrates, according to an exemplary embodiment, a side perspective view of a dissecting barrier unit in a vehicle blocking state.
(4) FIG. **2** schematically illustrates, according to an exemplary embodiment, a front view of a dissecting barrier unit in a vehicle blocking state.
(5) FIG. **3** schematically illustrates, according to an exemplary embodiment, a side view of a dissecting barrier unit in a vehicle blocking state.
(6) FIG. **4** schematically illustrates, according to an exemplary embodiment, a top view of a dissecting barrier unit in a vehicle blocking state.
(7) FIG. **5** schematically illustrates, according to an exemplary embodiment, a front view of a dissecting barrier unit in a vehicle blocking state, further comprising a stopper.
(8) FIG. **6** schematically illustrates, according to an exemplary embodiment, a side perspective view of a dissecting barrier unit in a vehicle passage state.
(9) FIG. **7** schematically illustrates, according to an exemplary embodiment, a front view of a dissecting barrier unit in a vehicle passage state.
(10) FIG. **8** schematically illustrates, according to an exemplary embodiment, a side view of a dissecting barrier unit in a vehicle passage state.
(11) FIG. **9** schematically illustrates, according to an exemplary embodiment, a top view of a dissecting barrier unit in a vehicle passage state.
(12) FIG. **10** schematically illustrates, according to an exemplary embodiment, a bottom view of a dissecting barrier unit.
(13) FIG. **11** schematically illustrates, according to an exemplary embodiment, a side perspective view of a dissecting barrier system in a vehicle blocking state.
(14) FIG. **12** schematically illustrates, according to an exemplary embodiment, a front view of a dissecting barrier system in a vehicle blocking state.
(15) FIG. **13** schematically illustrates, according to an exemplary embodiment, a top view of a dissecting barrier system in a vehicle blocking state.
(16) FIG. **14** schematically illustrates, according to an exemplary embodiment, a front perspective view of a dissecting barrier system in a vehicle passage state.
(17) FIG. **15** schematically illustrates, according to an exemplary embodiment, a side view of a dissecting barrier system in a vehicle passage state.
(18) FIG. **16** schematically illustrates, according to an exemplary embodiment, a top view of a dissecting barrier system in a vehicle passage state.
(19) FIG. **17** schematically illustrates, according to an exemplary embodiment, a side view of a vehicle approaching a dissecting barrier system in a vehicle blocking state.
(20) FIG. **18** schematically illustrates, according to an exemplary embodiment, a top view of a vehicle approaching a dissecting barrier system in a vehicle blocking state.
(21) FIG. **19** schematically illustrates, according to an exemplary embodiment, a front view of a vehicle approaching a dissecting barrier system in a vehicle blocking state.
(22) FIG. **20** schematically illustrates, according to an exemplary embodiment, a side view of a vehicle passing over a dissecting barrier system in a vehicle passage state.
(23) FIG. **21** schematically illustrates, according to an exemplary embodiment, a top view of a vehicle passing over a dissecting barrier system in a vehicle passage state.
DESCRIPTION OF THE PREFERRED EMBODIMENTS
(24) Before explaining at least one embodiment in detail, it is to be understood that the subject

matter is not limited in its application to the details of construction and the arrangement of the components set forth in the following description or illustrated in the drawings. The subject matter is capable of other embodiments or of being practiced or carried out in various ways. Also, it is to be understood that the phraseology and terminology employed herein is for the purpose of description and should not be regarded as limiting. In discussion of the various figures described herein below, like numbers refer to like parts. The drawings are generally not to scale.

(25) For clarity, non-essential elements were omitted from some of the drawings.

(26) The present subject matter is aimed at providing a dissecting element adapted to be used as barrier to approaching vehicles that are impacted by the elements that act as blades and dissect parts of the vehicle.

(27) The present subject matter is also aimed at providing a dissecting barrier unit, configured to be either in a vehicle blocking state, or a vehicle passage state. The dissecting barrier unit is further configured to allow free passage of pedestrians, animals and small vehicles like bicycles and motorcycles, in either state—vehicle passage state or vehicle blocking state.

(28) The present subject matter further provides a dissecting barrier system comprising a plurality of dissecting barrier units.

(29) Reference is now made to FIGS. **1-4** schematically illustrating, according to an exemplary embodiment, a different views of a dissecting barrier unit in a vehicle blocking state. According to one embodiment, a dissecting barrier unit **10** comprises a ramp **12** and a dissecting element **14** connected to the ramp **12**. According to another embodiment, the dissecting element **14** is pivotally connected to the ramp **12**. According to yet another embodiment, the dissecting element **14** is pivotally connected to the ramp **12** with a hinge **146**, illustrated for example in FIGS. **1** and **3**. The ramp can be portable.

(30) The dissecting element comprises a flat body having an upper side and a bottom side and opposite edges as well as a hinge on the bottom side wherein the hinge is connected to a surface. The dissecting element is configured to move between an erected state in which the flat body is in an upright position and a lying state in which the flat body is adjacently lying on the surface.

(31) According to one embodiment, the surface is a ramp **12** that is configured to be placed on a surface **500** (ground) in a place where there is a desire to block passage of vehicles. The ramp **12** is configured to be placed on any type of surface **500** known in the art, for example a paved road, a non-paved road, ground, sand, rock and the like. According to another embodiment, the ramp **12** is configured to be anchored to the surface **500**. Any mechanism known in the art for anchoring the ramp **12** to the surface **500** is under the scope of the present subject matter, for example screwing, nailing, pegging and the like. According to yet another embodiment, the ramp **12** comprises at least one anchoring element **122** configured to facilitate anchoring of the ramp **12** to the surface **500**. Any type of anchoring element **122** is under the scope of the present subject matter. An exemplary embodiment of an anchoring element **122** is, for example, an anchoring element **122** in a form of a hole in the ramp **12**, configured to accommodate a attaching element, for example a nail, a screw, a peg and the like, that is configured to be stuck in the surface through the hole. It is also possible that the nail or screw will be incorporated within the hole pointing downwardly towards the surface. It should be noted that even though the anchoring element **122**, which is a hole in the ramp **12** is part of the dissecting barrier unit **10**, according to some embodiments, the nail, screw, peg and the like are not necessarily part of the present subject matter and can be independently provided by a user of the dissecting barrier unit **10**. According to a further embodiment, the anchoring of the ramp **12** to the surface is permanent. According to yet a further embodiment, the anchoring of the ramp **12** to the surface is temporary, or in other words—reversible.

(32) According to one embodiment, the ramp **12** is configured to be connected to at least one other ramp **12**. Any mechanism known in the art for connecting the ramp **12** to at least one other ramp **12** is under the scope of the present subject matter, for example tying, connecting, welding and the like. According to another embodiment, the connection of the ramp **12** to at least one other ramp **12**

is permanent, for example by welding. According to yet another embodiment, the connection of the ramp **12** to at least one other ramp **12** is temporary, namely reversible. According to a further embodiment, the connection of the ramp **12** to at least one other ramp **12** is by using connectors, hooks and the like. According to an additional embodiment, the ramp **12** comprises at least one connecting element **124** configured to participate in the connection of the ramp **12** to at least one other ramp **12**. Any type of connecting element is under the scope of the present subject matter. According to the exemplary embodiment, illustrated for example in FIG. **1**, the connecting element **124** is a tube-like element attached preferably beneath the ramp **12** and configured to accommodate a shaft-like element (not seen) configured to be inserted into the tube-like element of one ramp **12** and a tube-like element of an adjacent ramp **12**, thus connecting the ramps **12** and holding them together.

(33) Even if there is another mechanism to connect the ramps to each other, the tubes that are discussed herein before, which are used for the connection (connecting elements **124**) of the ramps, are provided beneath the ramp **12** (especially in cases the ramp is curved, as will be discussed herein after) in order to strengthen the ramp onto which vehicles pass. This is clearly shown in FIG. **3** in which the connecting elements **124**, which are preferably elongated tubes that are attached beneath the ramp through its entire length. In FIG. **3** one can see that the two opposite tubes **124** are between the ramp **12** and the surface **500**.

(34) According to one embodiment, the ramp **12** is configured to allow passage of a vehicle over the ramp **12**. Thus, according to another embodiment, the ramp **12** is substantially flat, thus allowing passage of a vehicle over the ramp **12** while moving on the surface on which the ramp **12** is positioned. According to yet another embodiment, the ramp **12** is substantially curved as clearly seen in FIGS. **1** and **3**. According to still another embodiment, the degree of curvature of the ramp **12** is such that it allows free passage of a vehicle over the ramp **12**, for example like passing over a bump on a road, or less curved. It has been found by the inventor that a substantially curved shape confers strength to the ramp **12**, comparted to a substantially flat shape.

(35) According to one embodiment, the ramp **12** comprises at least one anchoring element **122** as described herein. According to another embodiment, the ramp **12** comprises at least one connecting element **124** as described herein. According to yet another embodiment, the ramp **12** comprises at least one anchoring element **122** and at least one connecting element **124**, as described herein.

(36) According to one embodiment, as described above, the dissecting barrier unit **10** comprises a ramp **12** and a dissecting element **14** connected to the ramp. The dissecting element **14** can be connected to any position on the ramp **12**. According to this embodiment, the ramp **12** has any structure and dimensions known in the art, for example a rectangular shape as illustrated for example in FIG. **4**.

(37) According to another embodiment, the ramp **12** can have a minimal size, enough only to connect to the dissecting element **14** and allow anchoring of the dissecting barrier unit **10** to the surface as described herein, as well as optionally to be connected to another at least one dissecting barrier unit **10** as described herein.

(38) According to an additional embodiment, the dissecting barrier unit **10** comprises a dissecting element **14**. According to an additional embodiment, the dissecting element **14** is shaped and functions according to embodiments described herein. According to still an additional embodiment, the dissecting element **14** is configured to be anchored to a surface by mechanisms described herein in relation to the ramp **12**.

(39) According to one embodiment, the dissecting element **14** is configured to be in a vehicle blocking state as described herein. This embodiment relates to a dissecting barrier unit **10** comprising a dissecting element **14** and to a dissecting barrier unit **10** comprising a ramp **12** and a dissecting element **14** connected to the ramp **12**.

(40) According to another embodiment, the dissecting element **14** is configured to be in a vehicle blocking state, as illustrated in FIG. **1-5**. In other words, the dissecting element **14** is at least

partially erected relative to the surface on which the dissecting barrier unit **10** is positioned. Additional embodiments relating to the vehicle blocking state of the dissecting element **14** are described hereinafter.

(41) According to one embodiment, the dissecting element **14** is pivotally connected to the ramp **12**. According to another embodiment, the dissecting element **14** is configured to be in a vehicle blocking state, as illustrated in FIGS. **1-5**. According to yet another embodiment, the dissecting element **14** is configured to be in a vehicle passage state, as illustrated in FIGS. **6-9**.

(42) In the vehicle blocking state, the dissecting element **14** is at least partially erected, while in the vehicle passage state the dissecting element **14** substantially rests on the surface **500** or on the ramp **12**. Thus, in the vehicle blocking state, the dissecting element **14** is configured to block passage of a vehicle, while in the vehicle passage state the dissecting element **14** is configured to allow passage of a vehicle over the dissecting element **14** without being harmed.

(43) According to one embodiment, the dissecting element **14** comprises a substantially flat body having two wide edges **142** at the upper side of the dissecting element and two narrow edges **144** at the bottom, while the narrow edges are longer than the wide edges. According to another embodiment, in the vehicle blocking state, the dissecting element **14** is positioned with the edges **142** and **144** facing a potential moving vehicle. In FIG. **4**, illustrating a top view of the dissecting barrier unit **10** in a vehicle blocking position, arrows **502** indicate possible directions of movement of a vehicle in relation to the orientation of the dissecting element **14** in which the action of the barrier is most effective. It can be seen in FIG. **4** that the edges of the dissecting element **14** face the directions indicated by arrows **502** of a vehicle. Thus, when a vehicle moves straight toward the erected dissecting element **14** in the vehicle blocking state, in direction **502**, at least one of the wide and narrow edges **142** and **144** of the dissecting element **14** is configured to impact the vehicle and block its movement. In addition, the dissecting element **14** is configured to potentially dissect portions of the vehicle because at least one of the edges, the narrow edges **144** and the wide edges **142** is the part of the dissecting element **14** that impacts the vehicle that moves towards the barrier in relatively high speed. In other words, according to one embodiment, at least one of the narrow edges **144** of the dissecting element **14** is configured to dissect a vehicle when the dissecting element **14** impacts the vehicle.

(44) FIG. **4** further illustrates by arrows **502** possibilities of the direction of movement of a vehicle toward the dissecting element **14**. Even when the direction of movement of the vehicle is not straight toward the dissecting element **14** (as the direction of the arrow in the middle), but rather angular, the dissecting element **14** is still configured to block the movement of the vehicle, and potentially dissect at least some portions of the vehicle.

(45) According to one embodiment, at least one of the edges **142** and **144** of the dissecting element **14** and at least one of the opposite edges is sharp, having a blade-like shape. As an example shown in FIG. **3**, a first narrow edge **144***a* is sharp having a blade-like shape; or a second narrow edge **144***b* is sharp having a blade-like shape; or both the first narrow edge **144***a* and the second narrow edge **144***b* are sharp and are having a blade-like shape.

(46) As mentioned above, in the vehicle blocking state, the dissecting element **14** is configured to erect substantially perpendicularly to the surface on which the dissecting barrier unit **10** is placed. This may be achieved, for example, by the shape of the hinge **146** with which the dissecting element **14** is pivotally connected to the ramp **12**. Alternatively or additionally, according to one embodiment, the dissecting barrier unit **10** comprises a stopper **16** configured to stop movement of the dissecting element **14** and hold the dissecting element **14** in a substantially perpendicular and firm erected position. The stopper **16** is illustrated in FIG. **5**.

(47) As mentioned herein before, FIGS. **6-9** schematically illustrates, according to an exemplary embodiment, different views of a dissecting barrier unit in a vehicle passage state. The structure of the dissecting element is shown in a non-erected state in which the dissecting element **14** is adjacently lying on the ramp **12**.

(48) Reference is now made to FIG. **10** schematically illustrating, according to an exemplary embodiment, a bottom view of a dissecting barrier unit. The holes of the anchoring elements **122** are shown to indicate that the holes pass through the ramp **12**. The bottom portion of the hinge **146** that allows the erection of the dissection element **14** is shown where the base of the dissecting element (not shown in the figure) is positioned.
(49) The present subject matter further provides a dissecting barrier system. Reference is now made to FIGS. **11-13** schematically illustrating, according to an exemplary embodiment, different views of a dissecting barrier system in a vehicle blocking state. According to one embodiment, a dissecting barrier system **1** comprises a plurality of dissecting barrier units **10**. Furthermore, in an embodiment in which the dissecting barrier system **1** comprises a plurality of dissecting barrier units **10**, at least some of the dissecting barrier units **10** are connected one to the other. According to another embodiment, all the dissecting barrier units **10** of the dissecting barrier system **1** are connected one to the other. According to yet another embodiment, the dissecting barrier units **10** of the dissecting barrier system **1** are not connected one to the other but are positioned one aside the other.
(50) It should be noted that the dissecting barrier system can also provide a plurality of dissecting elements that are directly connected to the surface—the ground.
(51) All the embodiments in relation to the dissecting barrier unit **10** pertain also to the dissecting barrier units **10** of the dissecting barrier system **1**.
(52) Reference is now made to FIG. **14-16** schematically illustrating, according to an exemplary embodiment, a front perspective view of a dissecting barrier system in a vehicle passage state. According to this embodiment, all the dissecting units **10** of the dissecting barrier system **1** are in a vehicle passage state. According to yet another embodiment, some of the dissecting barrier units of the dissecting barrier system are in a vehicle blocking state, and some dissecting barrier units are in a vehicle passage state. In other words, according to an additional embodiment, the dissecting barrier units of the dissecting barrier system **1** are independent one in relation to the other in terms of their state—vehicle blocking state and vehicle passage state. According to yet an additional embodiment, the dissecting barrier units **10** of the dissecting barrier system **1** depend one on the other in terms of their state—vehicle blocking state and vehicle passage state. In other words, the dissecting barrier system comprises, according to a further embodiment, a mechanism that controls the state of at least some of the dissecting barrier units **10**, for example by connecting wires that are connected to the dissecting element **14** of the dissecting barrier system **1** and allow simultaneous shifting of the dissecting elements **14** between the vehicle blocking state and the vehicle passage stage.
(53) As discussed herein before regarding the dissecting unit, the placement of the barrier for blocking vehicles is dependent on the direction of the vehicles. Reference is now made to FIG. **17** schematically illustrating, according to an exemplary embodiment, a side view of a vehicle approaching a dissecting barrier system in a vehicle blocking state. The barrier **1** is placed with one of the edges of the dissecting element **14** towards the direction of the coming vehicles. Vehicle **900** is shown to approach the barrier **1** in a position that is most effective as to the blockage of the vehicle. FIGS. **18** and **19** illustrates similar positioning of the vehicle versus the barrier, in different views.
(54) As illustrated in FIG. **18**, a length of a gap between two adjacent dissecting elements **14**, in a vehicle blocking position, which are part of a dissecting barrier system **1**, is defined as gap that is indicated by line **300**. In addition, a vehicle **900** approaching the dissecting barrier system **1** has a width indicated by line **950**. According to one embodiment, the gap **300** between two adjacent dissecting elements **14** in a vehicle blocking position, is smaller than the width **950** of the vehicle **900** (or any other typical vehicle) approaching the dissecting barrier system **1**. Therefore, the dissecting barrier system **1** is configured to block passage of vehicles **900**, because a vehicle **900** is not capable of passing through the gap **300** between two adjacent dissecting elements **14** in a

barrier blocking state. Nevertheless, the gap **300** between two adjacent dissecting elements **14** in a barrier blocking state allows free passage of pedestrians, animals and small vehicles like bicycles and motorcycles (not shown). It should be noted that even if the width of the vehicle is just a little smaller than the gap between the dissecting elements, it is most likely that an attempt to pass through two adjacent dissecting elements will receipt in at least dissection of the vehicle's wheels.
(55) Therefore, a method of blocking vehicles is provided in accordance with another embodiment of the present subject matter, while the method comprising: providing a dissecting element or a dissecting barrier unit as shown herein although the description, and positioning the dissecting element with the opposite edges directed towards a direction from which the vehicles are approaching.
(56) Reference is now made to FIGS. **20** and **21** schematically illustrating, according to an exemplary embodiment, a side view and a top view, respectively, of a vehicle passing over a dissecting barrier system in a vehicle passage state. According to one embodiment, when the dissecting elements **14** of a dissecting barrier unit **1** are in the vehicle passage state, passage of vehicle **900** over the dissecting barrier system **1** is allowed and possible without harming the vehicle. Needless to say that passage of pedestrians, animals and small vehicles like bicycles and motorcycles is also allowed when the dissecting elements **14** of the dissecting barrier unit **1** are in a vehicle passage state as in the blocking state.
(57) It should be noticed that the changes of the states—passing vehicles or blockage of vehicles—is controlled automatically or manually by changing the state of the hinges between the dissecting elements and the ramps. The movement of the dissecting elements from an upright position or erected position to lying position can be controlled electronically, manually, hydraulically, pneumatically etc. In any case in which the barrier is automatically controlled, it can be operated from a distance by a remote control.
(58) The subject matter describes a method to prevent passage of vehicles. The method of blocking vehicles comprises providing a plurality of dissecting element or a dissecting barrier units and positioning the dissecting element or the units with the opposite edges of the dissecting elements so as to be directed towards a direction from which the vehicles are approaching.
(59) It is appreciated that certain features of the subject matter, which are, for clarity, described in the context of separate embodiments, may also be provided in combination in a single embodiment. Conversely, various features of the subject matter, which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable sub combination.
(60) Although the subject matter has been described in conjunction with specific embodiments thereof, it is evident that many alternatives, modifications and variations will be apparent to those skilled in the art. Accordingly, it is intended to embrace all such alternatives, modifications and variations that fall within the spirit and broad scope of the appended claims.

Claims

1. A dissecting barrier unit for blocking a fast-approaching vehicle comprising: at least one dissecting element sized to reach at least the height of a chassis of the vehicle having a flat body with opposite side edges, wherein at least one edge of the opposite side edges is a sharp edge that is arranged to substantially face a potential direction of the vehicle; and a ramp onto which said at least one dissecting unit is pivotally connected, wherein the at least one dissecting element is configured to move between an erected active state in which the flat body is in an upright position capable of dissecting parts of the fast-approaching vehicle in front of wheels of the vehicle that first encounter the sharp edge and a horizontal inactive state in which the flat body is adjacently lying on the ramp allowing the vehicle to pass through, and wherein the dissecting element is configured to block the fast-approaching vehicle by dissecting the chassis, motor, and tires of the vehicle when the vehicle impacts the dissecting element.

2. The dissecting barrier unit of claim 1, wherein the at least one dissecting element is pivotally connected to the ramp by a hinge.
3. The dissecting barrier unit of claim 1, wherein the ramp is configured to permanently or reversibly be connected to the at least one other ramp.
4. The dissecting barrier unit of claim 1, wherein the ramp is configured to allow passage of a vehicle over the ramp when the dissecting elements are in a horizontal state.
5. The dissecting barrier unit of claim 1, wherein the ramp is substantially flat or curved.
6. The dissecting barrier unit of claim 1, wherein the flat body has an upper side that is wide and lower side that is relatively narrow.
7. The dissecting barrier unit of claim 1, wherein at least part of the edges of the dissecting element is sharp, having a blade-like shape.
8. The dissecting barrier unit of claim 1, wherein the dissecting element further comprises a stopper configured to stop movement of the dissecting element and firmly hold the dissecting element in the erected state.
9. The dissecting barrier unit of claim 1, wherein the ramp is configured to be anchored to a surface.
10. The dissecting barrier unit of claim 9, wherein the ramp is permanently anchored to the surface.
11. The dissecting barrier unit of claim 9, wherein the ramp is reversibly anchored to the surface.
12. The dissecting barrier unit of claim 1, wherein the ramp further comprises at least one connecting element for connecting the ramp to the at least one other ramp.
13. The dissecting barrier unit of claim 12, wherein the connecting element is a tube-like element attached beneath the ramp and configured to accommodate a shaft-like element configured to be inserted into the tube-like element of one ramp and a tube-like element of an adjacent ramp, thus connecting the ramps and holding them together.
14. The dissecting barrier unit of claim 5, wherein a degree of curvature of the ramp is such that it allows free passage of a vehicle over the ramp when the dissecting elements are in a horizontal state.
15. A dissecting barrier system comprising a plurality of dissecting barrier units according to claim 1.
16. The dissecting barrier system of claim 15, wherein at least two of the dissecting barrier units are connected one to the other.
17. The dissecting barrier system of claim 15, wherein each one of the dissecting barrier units is independently controlled.
18. The dissecting barrier system of claim 15, wherein each one of the dissecting barrier units is adapted to be in an erected state or in horizontal state.
19. The dissecting barrier system of claim 15, wherein each one of the dissecting barrier units is independently controlled.
20. The dissecting barrier system of claim 19, wherein the dissecting barrier units are remotely controlled.
21. The dissecting barrier system of claim 17, wherein the dissecting elements are remotely controlled.